Guide QC and analysis pipeline

Here is an overview of the actions needed before starting the QC and analyses, tips on installation, and a description of each script.

If you have questions, please contact kira.hoeffler@uib.no or anne-kristin.stavrum@uib.no.

Running the pipeline

1. Unzip the "Cohort_analysis_pipeline.zip" file to your file system at the location you intend to use as your working directory.
    - **Important**: Ensure that the 1_Install_packages, 2_Paths, 3_Run_pipeline, and the script folder are in the working directory."

Name	Date modified	Type	Size
Resources	19.06.2024 14:06	File folder	
script	19.06.2024 14:06	File folder	
1_Install_packages	19.06.2024 16:37	R File	4 KB
2_Paths	19.06.2024 15:44	R File	2 KB
3_Run_pipeline	19.06.2024 16:16	R File	4 KB

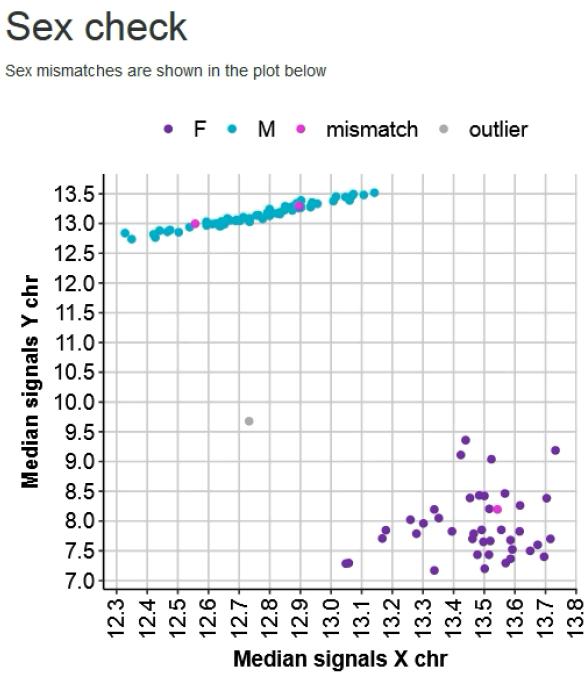
2. **Open script 1_install_packages.R**
    a. Update the **array.type** to one of "EPICv1", "EPICv2" or "450K"
    b. Run the code lines. If you experience **any issues with the installation,** please consult the installation guide further down in this document.
3. Please **save your samplesheet in an Excel file, with the following columns:**
    - AMP_Plate: This is the amplification plate used during the PCR process to amplify DNA samples for further analysis.
    - Sentrix_ID: A unique identifier for the microarray slide used in Illumina's Infinium methylation assay.
    - Sentrix_Position: The specific position on the Sentrix array slide where a particular DNA sample is placed.
    - Sex – coded as "M" or "F"
    - Age
    - Ethnicity (= genetic ancestry in this case, can be self-reported or preferably genetically predicted)
    - Case_Control – coded as "Case" or "Control"
    - *If multiple samples per individual, e.g. because of longitudinal analysis:* Baseline_sample with entry "Y" for all samples that should be included in the analysis
        - Alternatively, you can make sure that the sample sheet contains only one time point (Baseline) per individual.

Complete data is expected (if Ethnicity is unknown, please change it to "unknown"). **It is crucial to ensure that the sample sheet is accurately coded with the correct column names and proper coding for Sex and Case_Control to run the pipeline.**

We will generate a column called Basename using "SentrixID_SentrixPosition" for the sample names if this column is not already present in your sample sheet.

4. **Update the information in 2_Paths.** You do not need to run the 2_Paths.R script, just save your changes.
    a. Optional step: Use this only if you have expert knowledge & after consulting with us:
        i. We provide the option to define additional samples in this script that you want to exclude, e.g., if there is a known genotyping mismatch.
        ii. Add the sample's Basename (paste0("Sentrix_ID" "_" "Sentrix_Position")) to the additional_outliers vector and a reason for exclusion, e.g., "genotyping_mismatch," to the reason_exclusion vector.".
5. **Open the 3_Run_pipeline.R script** to see an overview of the scripts and the order in which they should be run.
    - A description of each script follows further down in this guide.
    - The pipeline is divided into four parts (plus the setup), and an HTML report can be generated after each part, providing an overview of the plots generated.
    - Please pay attention to warning messages when running the pipeline.

    a. Adapt the path to the 2_Paths.R script in the source() function at the start of the script.
    b. The code chunk "SET PATHS" must be run every time you return to running the pipeline after closing it.
    c. Run the first part of the pipeline under the header "PART ONE – SET-UP".
    d. Run the second part of the pipeline under the header "PART TWO – QC and Filtering pipeline I"
        i. **Important:** Check if clear outliers have been correctly identified ("grey dots") in the plot in the generated "Report_Part2_QC_Sex.html" in your working directory.
        ii. <u>If the outliers have not been correctly identified:</u>
            1. Identify which X and Y coordinates in the sex check plot can be used to separate the outliers from the main female and male clusters. A larger version of the plot can be found in output/Figures/Sex_plot_mismatch. Here is an example of how it looks when the outlier in grey is correctly identified:



2. if the default axes coordinates of x < 10 and y < 11 do not separate your outliers from the male and female clusters, please open the "2_Sex_check" script in the script folder and adapt the following line in the middle of the script (example in the box below):

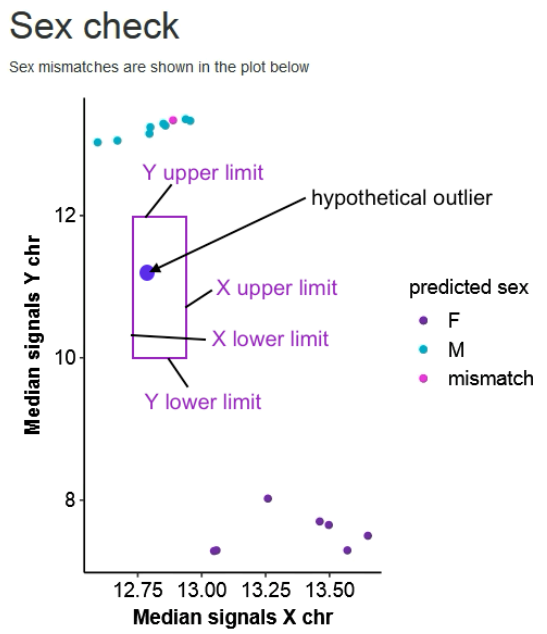
```
# SEX OUTLIER (TRESHOLDS SET BASED ON PLOT)
sex_outlier <- sex_df[which(sex_df$xMed < 10 & sex_df$xMed > 0 & sex_df$yMed < 11 & sex_df$yMed > 0), ]
```

Save your changes and return to the main "3_Run_pipeline" script. Re-run the code to run "2_Sex_check.R" and generate the report:

```
source("script/2_Sex_check.R")

# CREATE A REPORT
source("2_Paths.R")
rmarkdown::render("script/Report_Part1.Rmd",
                  output_file = paste0(dir_gen,"Report_Part1_QC_Sex.html"),
                  params = list(dir_gen = dir_gen))
```

We are happy to help if you are struggling with this step.



Sex_outlier <- sex_df[which(sex_df$xMed < X_UPPER_LIMIT & sex_df$xMed > X_LOWER_LIMIT & sex_df$yMed < Y_UPPER_LIMIT & sex_df$yMed > Y_LOWER_LIMIT),]

In this example, the purple box separates the outlier from the male and female clusters. Add the X and Y values that form the border of the box separating your outlier from other samples.

e. You can run Part 3 of the pipeline, "PART THREE – QC and Filtering Pipeline II," once the sex outliers have been correctly identified.
   i. **Important:** Check if all obvious outliers have been removed in the last three density plots (the ones with the grey background) in "Report_Part3_QC.html" in the working directory. Some variation on the X chromosome in females can be expected.

      <u>If that is not the case:</u>

      1. Similar to the sex outlier example, select X and Y coordinates that separate the outlier from the other samples.
      2. Open the respective script in the "script" folder (4_Filtering_autosomes, 5_Filtering_X_Females, or 6_Filtering_XY_Males).
      3. Go to the outlier section at the end of the script:

```
### OUTLIERS ###

# if there is an outlier in the filtered beta distribution plot, select an X value,
# and Y values that separate the outlier from the other samples. Run the code below to identify the outlier:

# RUN THIS CODE (REMOVE THE # AT THE START OF THE NEXT TWO LINES)
#outlier <- dens_df_long$Sample[which(dens_df_long$X == x_value & dens_df_long$value <= y_upper_limit & dens_df_long$value >= y_lower_limit)]
#outlier
```

      4. Add values for x_value, y_upper_limit, and y_lower_limit that separate the outlier from the other samples in this code and remove the hashtags.

```
outlier <- dens_df_long$Sample[which(dens_df_long$X == x_value & dens_df_long$value <= y_upper_limit & dens_df_long$value >= y_lower_limit)]
print(outlier)
```

      5. Re-run the respective script. The outliers will be printed at the end.

6. Add the names of the outliers to the outlier vector at the start of the respective script, e.g., outlier <- c("sample_name1", "sample_name2"):

```
### OUTLIER SELECTION (ADD OUTLIER WHEN THERE IS STILL AN OUTLIER AFTER RUNNING THE SCRIPT FOR THE FIRST TIME, SEE GUIDELINES) ###

outlier <- c() #identify outlier at the end of the script
```

7. Re-run the script once more.
8. Re-run the report generation via the "3_Run_pipeline.R" script.

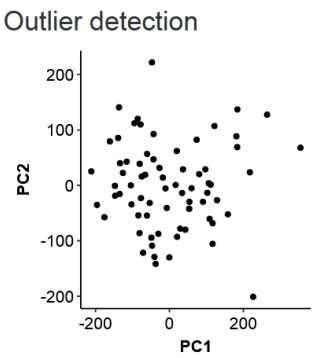
```
# CREATE A REPORT
source("2_Paths.R")
rmarkdown::render("script/Report_Part2.Rmd",
                  output_file = paste0(dir_gen, "Report_Part3_QC.html"),
                  params = list(dir_gen = dir_gen))
```

9. Feel free to contact us if you need assistance removing outliers that were not automatically removed by the pipeline.

f. You can run the fourth part of your pipeline, "PART FOUR – Normalization, Imputation, and Combining Autosomal and Sex Chromosome Data Sets," if all obvious outliers have been removed in "Report_Part3_QC.html".

   i. If your cohort consists of more than one genetic ancestry group, check the 3D ancestry plot ("Ancestry_PCs.html" in your working directory; open it in Firefox) to see if the clusters match the ethnicity in your sample sheet.

   ii. **Important:** Check if there are any remaining obvious outliers in the outlier detection PCA plot in "Report_Part3_Normalisation_Imputation_PCA.html.



<u>In the unlikely case that there are outliers:</u>

1. Open the "Normalisation_imputation_autosomes" script in the script folder and go to the end of the script.
2. Adapt the following lines by specifying which PC (PC1 or PC2) separates the outlier from the other samples and the value of this PC that separates the outlier from the other samples. Remove the hashtags.

```
# ADAPT AND RUN THIS CODE (REMOVE THE # AT THE START OF THE NEXT LINE), CHANGE WHICH PC SEPARATES THE OUTLIER AFTER THE $ SYMBOL AND THE NUMBER
outlier <- rownames(pca_scores[which(pca_scores$PC2 > 200), ])
print(outlier)
```

3. Re-run the entire script. It should print the outlier sample names at the end.
4. Open the "4_Filtering_autosomes" script in the script folder and add the outlier samples to the outlier vector at the start of the script, e.g.,:

```
### OUTLIER SELECTION (ADD OUTLIER WHEN THERE IS STILL AN OUTLIER AFTER RUNNING THE SCRIPT FOR THE FIRST TIME, SEE GUIDELINES) ###

outlier <- c("9741779074_R04C02", "9904796179_R05C02") # identify outlier at the end of the script
```

5. Save the script.
6. Re-run PART THREE of the pipeline from source("script/4_Filtering_autosomes.R") and then the entire PART FOUR.

g. You can run the fifth part of the pipeline, "PART FIVE – Analysis," when there are no remaining outliers in the outlier detection PCA plot in "Report_Part3_Normalisation_Imputation_PCA.html."

h. Please share the generated reports and the Ancestry_PCs file (all HTML files in the working directory) with us:

- We will choose the model with a lambda value close to 1.0.
  - If your QQ plots are all inflated, we might go back to see if we can further reduce the inflation, e.g., by adding SVs.
- VIF values above 3 indicate multicollinearity issues. Control probe PCs with VIF > 10 are automatically excluded from the analysis.

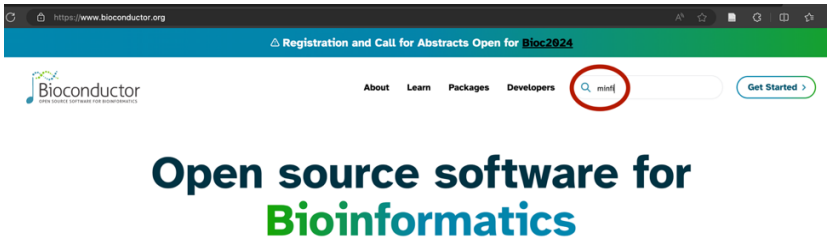
Installation guide

Two reasons why the **installation of new packages might not work** for you are:

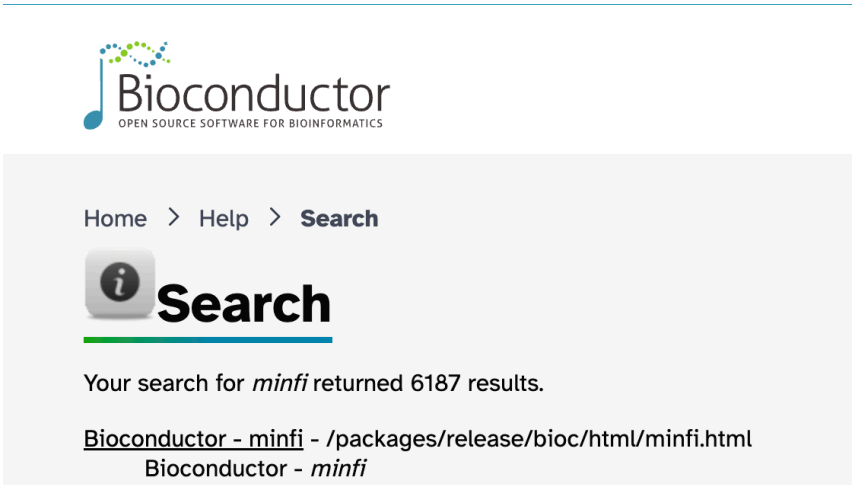
1. The functions for installing from Bioconductor or GitHub in the script will not work if you do **not have an internet connection** on your secure server. In that case, please install the packages manually (see below).
2. You will encounter errors when it is **not possible to automatically install some of the dependencies** on your system. This means that the package you are trying to install first requires its dependencies to be installed. The error message will indicate which dependencies are missing. Please install these packages manually (see below) if the standard functions like install.packages() and BiocManager::install() do not work for you.

Install manually from Bioconductor

1. Search for the package name on the Bioconductor website:
   [Bioconductor - Home]



2. Select the first hit



3. Download the source package (at the bottom of the website):

Package Archives

Follow Installation instructions to use this package in your R session.

Source Package	minfi_1.48.0.tar.gz

4. Upload the package folder to your server
5. In Rstudio,
    a. Define your package path:
    package_path <- "Path_to_folder"
    b. Install the package:
    install.packages(package_path, repos = NULL, type = "source")

Install manually from Github

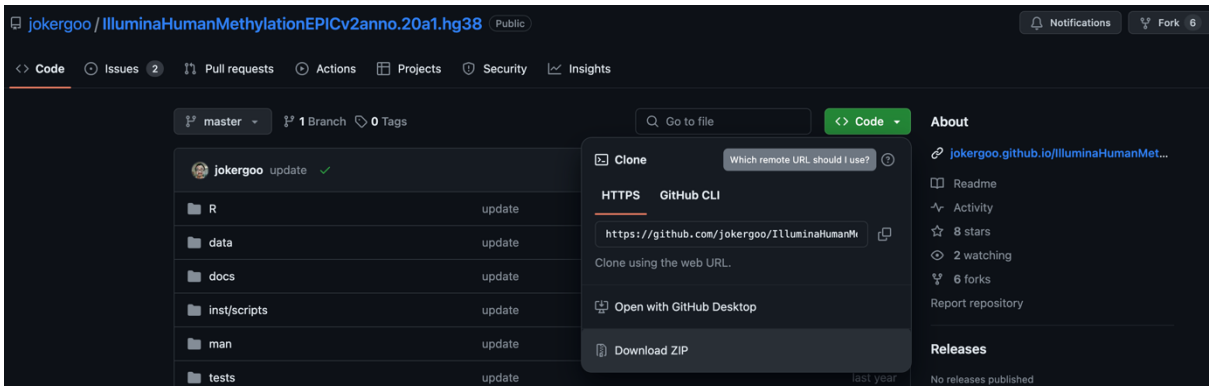
You will need the following three packages for the EPICv2 pipeline:

GitHub - schalkwyk/wateRmelon: Illumina 450 and EPIC methylation array normalization and metrics

GitHub - jokergoo/IlluminaHumanMethylationEPICv2manifest

GitHub - jokergoo/IlluminaHumanMethylationEPICv2anno.20a1.hg38

1. Open the link to the Github package (see above)
2. Go to **Code** and select **download ZIP** to download the package:



3. Upload the zip folder to your server
4. Unpack the zip folder / extract
5. In Rstudio,
    a. Define your package path:
    package_path <- "Path_to_folder"
    b. Install the package:
    install.packages(package_path, repos = NULL, type = "source")

Overview of the scripts

Here you will find further information on the scripts if you are interested in the details.

0_Setup

Some basic testing is performed to check the sample sheet, the information given in paths, and the installation of all needed packages. The folder structure for the output folder is set up. ggplot2 themes are defined. The sample sheet is copied to output/Tables. During the QC process, information will be added to the copied sample sheet.

1_IDAT_to_RGset

An RGset is generated from raw IDAT files and background-corrected using negative control probes. The bead counts per probe are extracted, and probes where >5% of samples have bead counts <3 are identified for later exclusion. Detection p-values are calculated.

2_Sex_check

Sex is predicted from DNA methylation values (from the RGset), and samples with a mismatch between predicted and reported sex are flagged. Clear sex outliers are also flagged for later exclusion.

3_RGset_to_intensities

The rs SNP probe betas are extracted from the RGset and saved for later ancestry prediction. Bisulfite conversion is confirmed, and samples that do not reach 90% bisulfite conversion are flagged. The intensities of the normal probes and positive control probes are extracted from the RGset, and minfi intensities are calculated based on the RGset. Probes with median intensity values < 10.5 are flagged.

If you have blood samples, blood cell proportions are estimated using the estimateCellCounts2 function based on a reference dataset ("FlowSorted.Blood.EPIC.Robj) provided in the "Resources" folder. The code for EPICv2 is improvised here as it has not been updated yet for the new array. The 450K cell type proportions are estimated based on the EPIC reference dataset as recommended in this communication: estimateCellCounts2 error in match.arg(referencePlatform) · Issue #10 · immunomethylomics/FlowSorted.Blood.EPIC · GitHub.

4_Filtering_autosomes

The raw beta and M value distributions are visualized, and the probes are filtered on autosomes. The detection p-value threshold is applied, setting all values not reaching the threshold to NA. Samples with sex mismatch or those that are outliers in the sex plot are excluded. Manually identified outliers (e.g., samples with genotype mismatches) are also excluded in this script.

Cross-reactive and SNP probes (10bp distance to the CpG site MAF 0.01) are excluded (PMID:23314698, PMID:27924034, PMID:27717381, PMID:38455390, PMID:38448820). Only the "0bpSNP probes" (overlapping with CpGs 0bp distance MAF 0.05) are kept separately for later

ancestry prediction. In addition we remove flagged probes and probes with mapping inaccuracies based on Illumina product files (Infinium MethylationEPIC v2.0 Product Files (illumina.com))

After the first filtering steps, beta values are used to calculate marker and sample call rates. Probes and samples with more than 5% missing data are excluded.

The filtered beta and M value distributions are visualized (separately for the 0bp SNP probes).

5_Filtering_X_Female

This script is similar to script 4, but it is specifically for X chromosomes in females.

6_Filtering_XY_Male

This script is similar to script 4, but it is specifically for X and Y chromosomes in males. A less strict detection p-value threshold is chosen as the intensities of the sex chromosomes in males are weaker than on the autosomes and X chromosomes in females.

7_Normalisation_imputation_autosomes

Two types of normalizations are applied on the autosomes: first quantile normalization of the six types of intensities, followed by BMIQ on the beta values. The distribution of NA values per probe is saved, and champ.impute (KNN) is used to impute missing beta values.

Replicated probes on the EPICv2 are handled similar to the rmPosReps function of the DMRcate package based on a publication by Peters et al. 2024 BMC Genomics (PMID:38448820). Precision was chosen over sensitivity and the weighted average was calculated instead of the normal average.

The distributions of the final beta and M values are visualized.

A PCA is run on the final M values, and the first PCs colored by CaseControl Status, Sex, Ethnicity, AMP plate, Sentrix position, and Sentrix ID are visualized.

If you have saliva samples, cell type proportions are estimates using HEpiDish (Zheng et al., 2018; PMID: 29693419).

8_Normalisation_imputation_X_female

The intensities of the X chromosome in females are quantile normalized, the distribution of missing values per probe is saved, champ.impute (KNN) is used to impute missing beta values, and the final beta value distribution of the X chromosome in females is visualized. For the EPICv2, replicated probes are handled (see script 7 for details).

9_Normalisation_imputation_XY_male

This script is similar to script 8 but for males, including the X and Y chromosomes.

10_Combined_file_X_Females

DNA methylation values from autosomes and X chromosomes are combined for the female samples, and beta and M value distributions are visualized.

11_Combined_file_XY_Males

DNA methylation values from autosomes, X, and Y chromosomes are combined for the male samples, and beta and M value distributions are visualized.

12_Combined_file_Females_Males

DNA methylation values of autosomes and X chromosomes are combined for females and males. Samples that did not pass the sex chromosome quality control are removed from the autosome files. A sample sheet containing only samples that passed the QC is saved.

13_Normalisation_imputation_0bpSNP_probes

The intensities of the 0bpSNP probes are quantile normalized, and missing values are imputed using champ.impute with KNN. Genotypes of the rs SNP probes are called. Cell type PCs, control probe PCs, and ancestry PCs are calculated, and scree plots (showing how much variation is explained by each PC) are generated.

For the ancestry PCs, the 0bpSNP probes are first residualized (adjusted for cell type PCs (5 for blood, two for saliva), Sex, Age, and three control probe PCs). The residualized 0bpSNPs are combined with the called rs SNP probes, a PCA is performed, and a 3D plot for the ancestry PCs is generated.

14_Run_analysis_limma

Sex-specific analyses will be run when there are at least 35 males and 35 females. Otherwise, sex-adjusted analyses will be run. Reports will be generated.

<u>FINAL CASE-CONTROL MODEL:</u> M values ~ CaseControl (+ Sex) + Age + smoking (M values of cg05575921 in AHRR) + cell type PCs (5 in blood, 2 in saliva) + control probe PCs + ancestry PCs.

Different combinations of the number of control probe PCs and ancestry PCs are tested. Control probe PCs that show multicollinearity issues (VIF > 10) are automatically removed from the model. The models are run, lambdas are extracted, and QQ plots are generated.